

RAI-818: INTELLIGENT **MOBILE ROBOTICS**

- **Introduction**
 - What is a Robot?
 - Historic Robots
 - Course Details
 - What is Motion Planning?

Text : Principles of Robot Motion – Theory, Algorithms and Implementations
H. Choset, K.M. Lynch, S. Hutchinson, G. Kantor, W. Burgard, et al.

INSTRUCTOR:
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Introduction



• What is a Robot?

What is a Robot?



I, Robot (2004)

Automaton
(Greek, autos \self" +
matos \thinking,
animated, willing")

- from English translation of 1920 play "Rossum's Universal Robots" by Karel Capek
- from Czech robotnik (slave), robota (forced labor, drudgery), robotiti (to work, drudge)
- from Slavic (arabeit) related to German Arbeit (work)
- Word coined by Capek's brother Josef, who used it initially in a short story
- Robotics coined in 1941 in a science fiction context by Isaac Asimov

Robotics Institute of America: re-programmable multi-functional manipulator designed to move materials, parts, tools, or specialized devices through variable programmed motions for the performance of a variety of tasks

Device that automatically performs complicated often repetitive tasks, or a mechanism guided by automatic controls



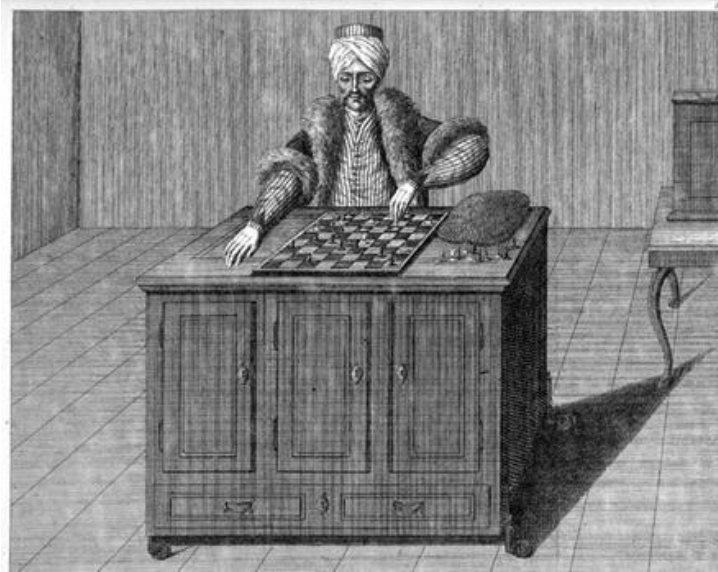
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Introduction

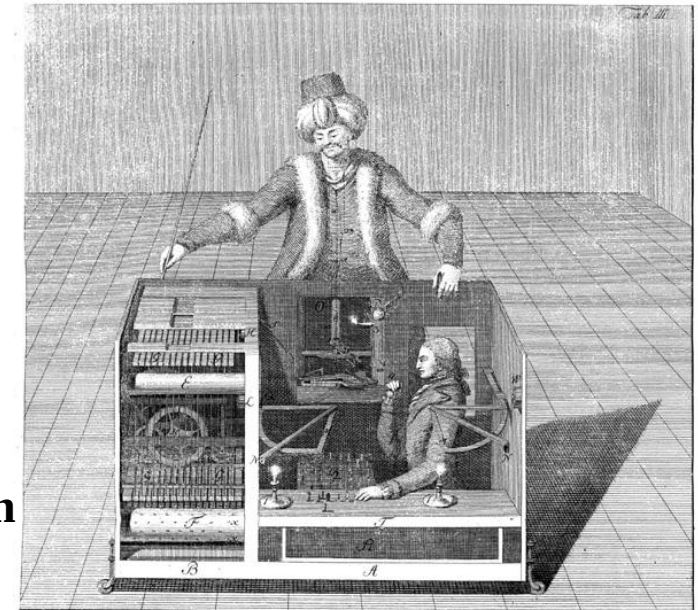


- What is a Robot?
- **Historic Robots**

First Robot - The Turk / Automaton Chess Player (1770)



- Constructed by Wolfgang von Kempelen in 1770
- Played many exhibition chess games
- Solved the knight-tour problem
- Even played against Benjamin Franklin in France



... it was a fake, however, human player hid inside machine



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- What is a Robot?
- **Historic Robots**

First Real Robot - Unimate(1961)



- Created by George Devol
- Worked on a General Motors assembly line in New Jersey in 1961
- Job consisted of transporting die castings from an assembly line and welding these parts on auto bodies
- Conducted into the Robot Hall of Fame in 2003



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Introduction



- What is a Robot?
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Resources

- **Textbooks:**

- **Principles of Robot Motion, Theory Algorithms and Implementations** *by Howie Choset, Kevin Lynch, Seth Hutchinson, George Kantor Wolfram Burgard, Lydia Kavraki, and Sebastian Thrun*
- **Introduction to Autonomous Mobile Robots** *by Roland Siegwart and Illah R.Nourbakhsh, The MIT Press, 2004*

- **Reference Books**

- **Robot Motion Planning**, *by Jean-Claude Latombe, Kluwer Academic Publishers, 1991.*
- **Probabilistic Robotics** *by Sebastian Thrun*
- **Planning Algorithms**, *by Steven, M, LaValle*
- **Robot Motion Planning** *by Jean Claude Latombe*
- **Computational Principles of Mobile Robotics**, *by Gregory Dudek and Michael Jenkin.*



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- What is a Robot?
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 - **Syllabus**

Syllabus

Part I: The Plan: Intelligent Motion Planning Algorithms

- Intelligent Bugs
- How to Plan Motions Intelligently within any Configuration Space
- Sampling-Based Motion Planning Techniques

Part II: The Robot: Probabilistic Localization for Autonomous Mobile Robots (AMRs)

- Fundamentals of Kinematics and Control for AMRs
- Fundamentals of Sensing Systems used on AMRs
- Bayesian Framework for Recursive State Estimation
- Probabilistic Localization with Kalman/Particle Filters

Part III: The World: Perception, Localization and Mapping for Intelligent Robot Sys

- Introduction to Simultaneous Localization and Mapping (SLAM)
- Perception and reasoning
- Gaussian Processes for interpreting sensor observations
- Feature-Based SLAM
- Feature-Less SLAM

Case Study: Implementation of Intelligent Mobile Robots Working in Real Environments



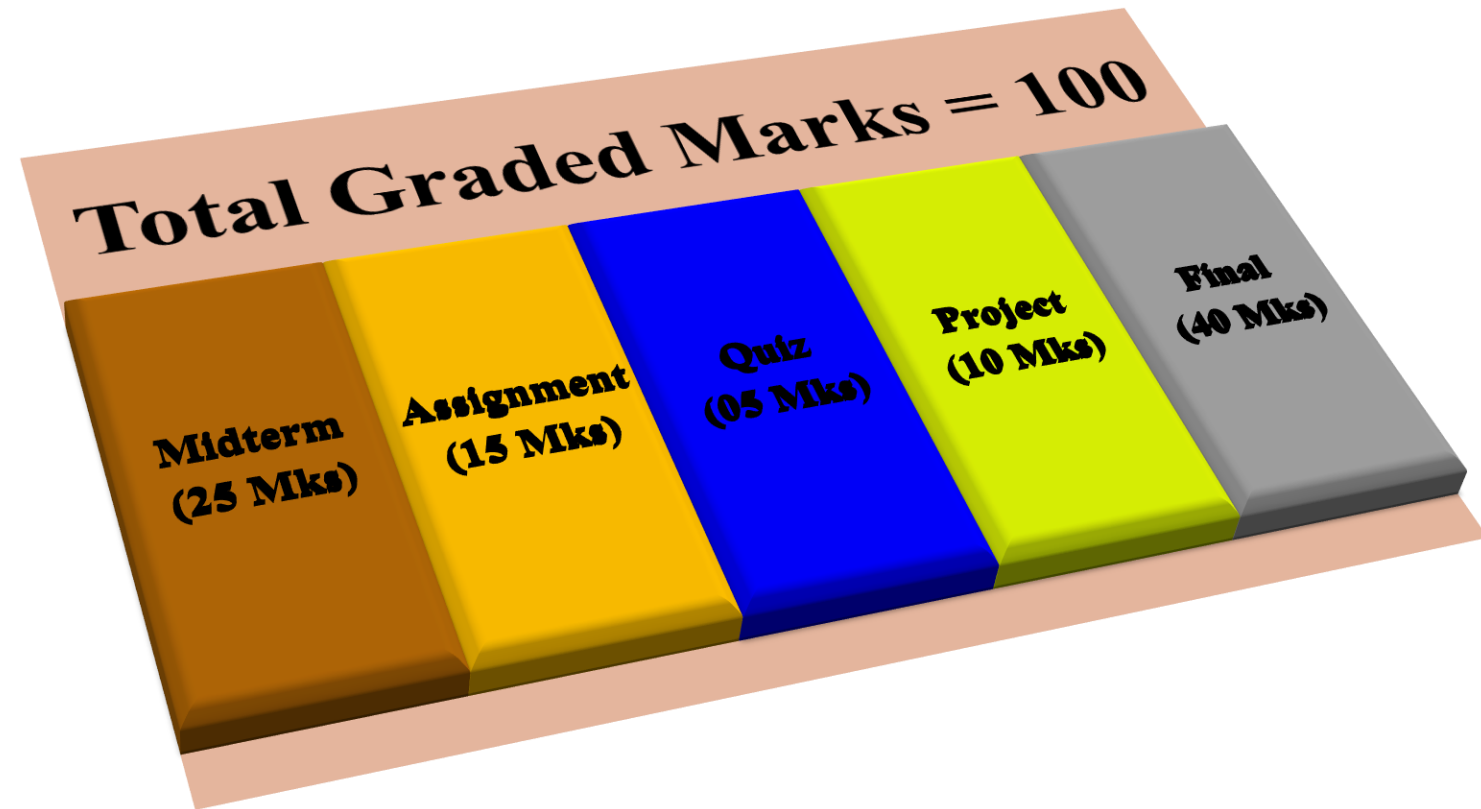
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- What is a Robot?
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 - Syllabus
 - **Course Load**

Course Load



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Course Load

- **How to submit assignments**
 - **Due at 11:59 pm on the due date (upload to LMS/MS Teams)**
 - **Worth 100 points (20 bonus points for best complete assignment)**



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 - **Focus**

Focus

- Two themes:
 - ***Motion Planning***: How can the robot automatically plan and execute a sequence of motions that avoids collision with obstacles and accomplishes the assigned task?
 - ***Localization and Mapping***: How can the robot use sensor-based information to determine its own state and model the world?
- Emphasis on algorithms, analysis, and implementations
- Illustrated with practical applications arising in diverse areas such as ***mobile systems, navigation and exploration, robot manipulation, computer animation, video games, computational biology, and medicine***



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 - **My Goals**

My Goals for the Class

- *Teach* people motion planning
- *Learn* about motion planning
 - From teaching the book
 - From class
- See impressive *demonstrations/simulations*
- *Identify* research areas in Motion Planning
- *Motivate* students towards further research in motion planning
- *HAVE FUN*



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- **Motion Planning**
 - **What is Motion Planning?**

What is Motion Planning?

- Determining where to go
...more than a search (or a geometric search)
- Motion planning is aimed at enabling robots with the capabilities of *automatically deciding and executing a sequence of motions* in order to *achieve a task without collision* with other objects in a given environment
- Basic Problem: *Static obstacles*
- Extended Problems: *Moving obstacles, Multiple robots, Dynamic Constraints, Planning with uncertainties, etc*



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 - What is Motion Planning?
 - **Trends**

Trends in Robotics/Motion Planning

Classical Robotics (mid-70's)

- exact models
- no sensing necessary

Reactive Paradigm (mid-80's)

- no models
- relies heavily on good sensing

Hybrids (since 90's)

- model-based at higher levels
- reactive at lower levels

Probabilistic Robotics (since mid-90's)

- seamless integration of models and sensing
- inaccurate models, inaccurate sensors



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 - **Mathematical Rigour**

Mathematical Rigour

Symbol	Meaning		
$\exists$	there exists	J	Jacobian
$\forall$	for all	Γ	Christoffel symbol
∞	infinity	RM	roadmap
$\in$	element	$\mathcal{W}$	workspace
$\notin$	not in	$\mathcal{Q}$	configuration space
s.t.	such that	$\mathcal{Q}_{\text{free}}$	free space
$\mathbb{R}$	real numbers	$x(k)$	state at time k
$\mathbb{R}^m$	m -dimensioned real numbers	$\ x\ $	norm of x
$\cup$	union	$\subseteq$	subset of
$\cap$	intersection	$\subset$	strict subset of
$\setminus$	set difference	$\text{cl}(A)$	closure of A
$\Rightarrow$	implies. $p \rightarrow q$ is p implies q	T^n	n -dimensional torus
$\Leftarrow$	implies. $q \rightarrow p$ is q implies p	S^n	n -dimensional sphere in $\mathbb{R}^{n+1}$
$\iff$	if and only if	$SO(n)$	special orthogonal group
S^1	a circle	$SE(n)$	special Euclidean group
∇	gradient	$B_\epsilon(q)$	open ball of radius ϵ centered at q
D	differential or distance to closest obstacle (depending context)	Df	differential of f
d_i	distance to obstacle i in either the workspace or configuration space (depending on context)	∇f	gradient of f
$d(x, y)$	distance between the two points x and y	∇	affine connection
Null	null space	$\nabla_{Y_1} Y_2$	covariant derivative of Y_2 with respect to Y_1
		C^0	continuous
		C^n	n times differentiable
		$\langle x, y \rangle$	inner product of x and y
		I	identity matrix
		$\text{atan2}(y, x)$	returns angle to (x, y) in the plane in range $[-\pi, \pi)$
		$T_x \mathcal{M}$	tangent space of $\mathcal{M}$ at x
		$T\mathcal{M}$	tangent bundle of $\mathcal{M}$
		$[f, g]$	Lie bracket of vector fields f, g
		$\overline{\text{Lie}}(\mathcal{G})$	the Lie algebra of a set of vector fields $\mathcal{G}$
		$\overline{\mathcal{D}}$	involutive closure of the distribution $\mathcal{D}$
		$\mathcal{U}_+$	control set positively spanning $\mathbb{R}^m$
		$\mathcal{U}_+$	control set spanning $\mathbb{R}^m$
		$\langle Y_1 : Y_2 \rangle$	the symmetric product of vector fields Y_1 and Y_2



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 - **Live Experiments**

Live Motion Planning Experiments

- *Person 1 walks through some obstacles*
- *Person 1, looking at Person 2, directs Person 2 through obstacles*
- *Person 1, looking at Person 2 with eyes closed, directs Person 2 through obstacles*
- *Person 1, looking at a map and not Person 2, whose eyes are still closed, directs Person 2 through obstacles*
- *Person 1, looking at an object and Person 2, whose eyes are closed, directs Person 2 to grab an obstacle*
- *How do we do the last experiment with a map?*



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What did we assume?

- *Perfect sensors?*
 - ♦ What information
 - ♦ Uncertainty
- *Perfect control?*
 - ♦ What controls?
 - ♦ Uncertainty
- *Perfect thinking?*
 - ♦ Knowledge of the world? Complete?
 - ♦ Processing the world? Everything?



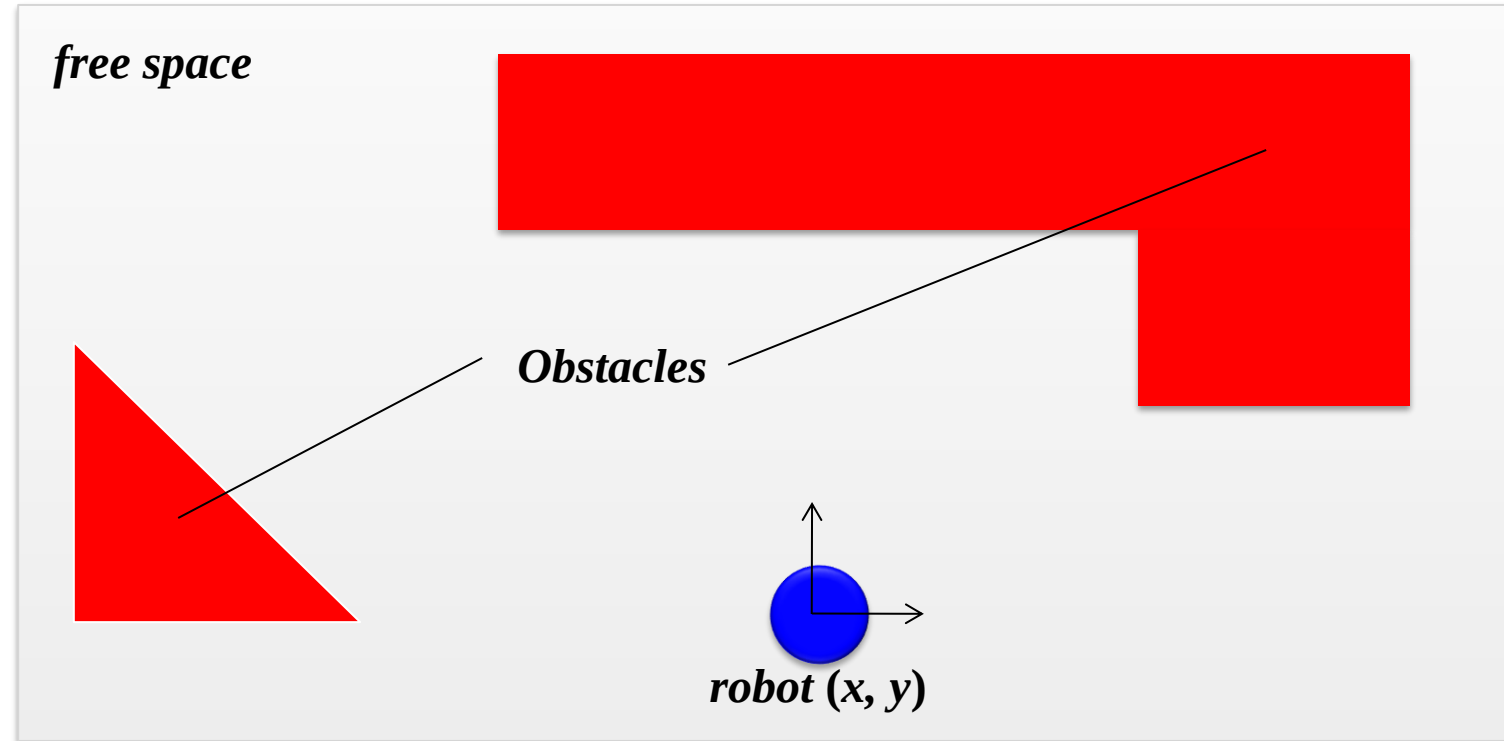
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 - **Examples**

Example of a World (and Robot)



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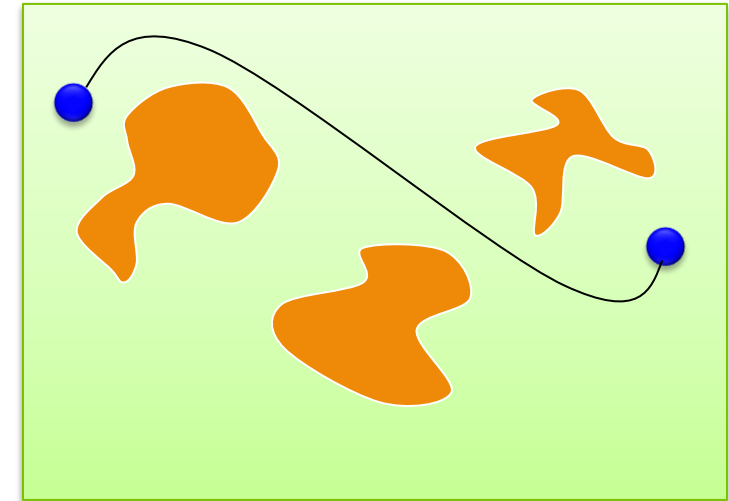
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Basic Path Planning

Problem Statement:

Compute a continuous sequence of collision-free robot configurations connecting the initial and goal configurations

2D Example



- Geometry of environment
- Geometry and kinematics of robot
- Initial and goal configurations



**Path
Planner**



**Collision-free
path**



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 - **Statement**

Motion Planning Statement

If W denotes the robot's workspace, and WO_i denotes the i 'th obstacle

Then the robot's free space, W_{free} , is defined as:

$$W_{free} = W - (\cup WO_i)$$

Motion planning occurs in the configuration or C -space, $R(q)$ denotes the set of points of the ambient space occupied by the robot at configuration q

And a path $c \in C^0$ is $c : [0,1] \rightarrow W_{free}$

where $c(0)$ is q_{start} and $c(1)$ is q_{goal}



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 - **Types**

Types of Motion Planning (1)

- **Static**
All information about obstacles is known a priori, and the motion of the robot is designed from the given information
- **Dynamic**
Only partial information is available about the obstacles e.g., the visible parts of the obstacles

The robot plans a path based on the available information

It discovers more obstacle information



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Types of Motion Planning (2)

- *Multimovers Problem*
more than one robot (e.g. Aibo team in RoboCup)
- *Conformable Problem*
objects can change shapes (e.g. Protein Folding, robot arms)
- *Nonconformable Problem*
objects that cannot change shapes (mobile robot on terrains)
- *Time-varying Problem*
environment or the configurations of obstacles change
- *Time-invariant Problem*
environment or the configurations of obstacles does not change
- *Movable-object Problem*
robots can move a subset of objects (e.g. Aibo team and PushPush)



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Types of Motion Planning (3)

- **Constrained**

velocity and acceleration limits

curvature of the robot's path

e.g. Trajectory Planning

- **Unconstrained**



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 - **Algorithms**

Types of Algorithms

- **Exact Algorithms**
either find a solution or prove that there is no solution
computationally expensive
- **Heuristic Algorithms**
generate a solution in a short time
may fail to find a solution for difficult problems
- **Global Algorithms**
take into account all the information in the environment, and plan a motion from the start to goal configuration
- **Local Algorithms**
designed to avoid obstacles in the vicinity of the robot and use information about nearby obstacles only



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 - Algorithms
 - **Basic Steps**

Basic Steps of Motion Planning

- Determine the *configuration parameters* of the robot in the given *configuration space*
- *Represent* the robot and objects properly
- Select a *motion planning approach* suitable to the motion planning problem at hand
- Select an *appropriate search method* to find a solution path
- *Optimize the solution* path for a shorter and smoother path



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 - Basic Steps
 - **Trends in Robotics Research**



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